

(2 Hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculators is **allowed**.

1. Attempt any two of the following: **12**

- a. Explain six types of foundational languages used for Artificial Intelligence briefly.
- b. Describe the Turing test, designed to provide an operational definition of Artificial Intelligence.
- c. Explain the PEAS description of the task environment for automated taxi and medical diagnosis systems.
- d. Define the following terms with examples:-
 - i) Agent
 - ii) Rational agent
 - iii) Irrational agent
 - iv) Learning agent
 - v) Omniscience agent
 - vi) Autonomy agent

2. Attempt any two of the following: **12**

- a. Describe the different components involved in problem formulation.
- b. Formulate 8-puzzle problem for the following scenario and also mention the steps and path cost required to reach goal state.

1	2	3
4	8	-----
7	6	5

Initial State

1	2	3
4	5	6
7	8	-----

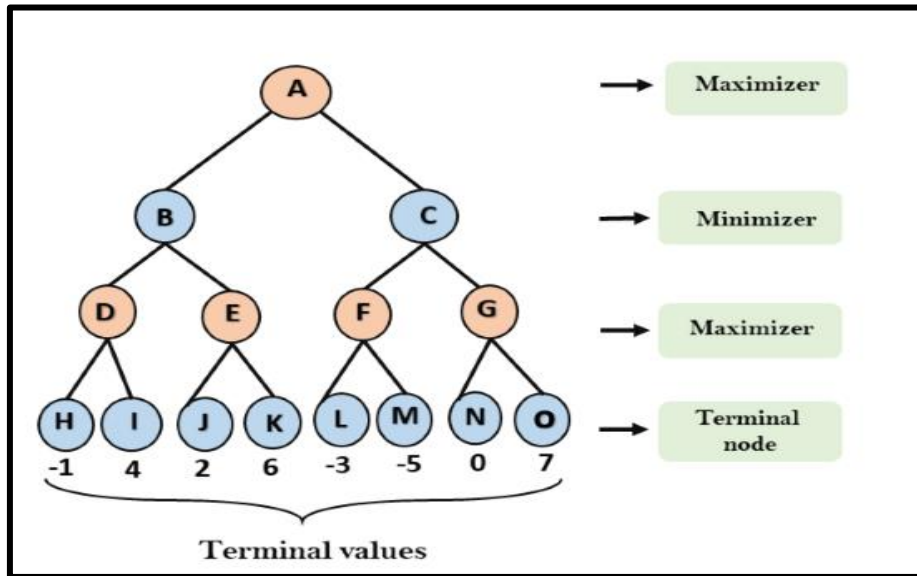
Goal State

- c. Provide an overview of Greedy First Search.
- d. Explain the State-space search diagram for the Hill Climbing algorithm in detail.

3. Attempt any two of the following: **12**

- a. Explain five state-of-the-art Game Programs.
- b. Explain and differentiate between two types of Inference systems for Knowledge-Based Agents (KBA).
- c. Describe five connectives of Propositional Logic and highlight its drawbacks.

- d. Describe the min-max algorithm and evaluate a given game tree using min-max to find the value of a node.



4. Attempt any two of the following:

12

- Define First Order Logic and explain its syntax and semantics.
- Describe the knowledge engineering process using First Order Logic in detail.
- Convert the below given Facts into FOL and Prove that "Colonel is a Criminal" using **Forward-Chaining**.
 - It is crime for an American to sell weapons to the enemy of America.
 - Country Nono is an enemy of America.
 - Nono has some Missiles.
 - All the Missiles were sold to Nono by Colonel.
 - Missile is a weapon.
 - Colonel is American.

- d. Compare and contrast Forward and Backward chaining reasoning methods.

5. Attempt any two of the following:

12

- Write a short note on:-
 - Total Ordered Planning
 - Partial Ordered Planning
- Explain two different types of algorithms used for planning as state space searches.
- Write short notes on Categories and Objects, and Mental Events and Mental Objects.
- Define reasoning systems for categories, explain Semantic Nets, and mention their advantages and disadvantages.